

MID TERM EXAMINATION

Paper: Mid Term Examination - Set B | Duration: 3 Hours | Max Marks: 53 | Date: 14-03-2026

Total Questions: 15 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. Which of the following operations is NOT typically associated with a standard Queue data structure? [2 Marks]
 - a) Enqueue
 - b) Dequeue
 - c) Pop
 - d) Front

2. In a circular linked list, what does the 'next' pointer of the last node point to? [2 Marks]
 - a) NULL
 - b) Itself
 - c) The first node of the list
 - d) The second node of the list

3. What is the main structural difference between an array and a record (struct/object) in terms of the types of data they can store? [2 Marks]

SECTION A: SHORT ANSWERS

4. What is the primary characteristic that distinguishes a linear data structure from a non-linear data structure? [2 Marks]

5. Which two primary operations define the behavior of a Queue data structure? [2 Marks]

SECTION A: MCQS

6. What is an Abstract Data Type (ADT)? [3 Marks]
 - a) A specific implementation of a data structure in a programming language.
 - b) A mathematical model that defines the data values and the operations on that data, without specifying their implementation.
 - c) A data structure that stores only abstract values or pointers.
 - d) A pre-defined class or library in a programming language.

7. If an algorithm has a time complexity of $O(n^2)$, and its input size n is doubled, how does its execution time approximately change in the worst case? [3 Marks]
 - a) It doubles.

- b) It quadruples.
- c) It remains the same.
- d) It halves.

8. When implementing a stack using a singly linked list, which end should new elements be pushed onto for $O(1)$ efficiency, assuming head points to the top of the stack? [3 Marks]

- a) The tail end of the list.
- b) The middle of the list.
- c) The head end of the list.
- d) It does not matter which end, as all operations are $O(1)$.

SECTION A: SHORT ANSWERS

9. In algorithm analysis, what does it signify if an algorithm has an $O(1)$ time complexity? [3 Marks]

10. A programmer needs to design a system where customer support requests are handled based on their severity level (e.g., critical, high, medium, low). Which variant of a queue would be most suitable for this scenario? [3 Marks]

11. Briefly explain the concept of Space Complexity in algorithm analysis. What are the two main types of space that contribute to it? [5 Marks]

12. Discuss two common applications where a Queue data structure is primarily used. [5 Marks]

SECTION A: MCQS

13. Describe the steps to delete a node with a specific value X from a singly linked list, assuming head is the pointer to the first node. [6 Marks]

14. Using a linear array-based implementation of a queue, describe how you would determine if the queue is full and if it is empty, assuming front and rear pointers are used and MAX_SIZE is the maximum capacity of the array. [6 Marks]

SECTION A: SHORT ANSWERS

15. Explain the difference between "Pass by Value" and "Pass by Reference" when passing a singly linked list (represented by its head pointer) to a function. Discuss the implications of each method on whether the function can modify the original list structure (e.g., change the head). [6 Marks]